

How to initialize the super class properties (non static field) using sub class object :

In order to initialize the super class properties (super class non static fields) through sub class object, we should use super keyword.

What is super keyword :

* super is a keyword in java which is used refer the **super class members OR super class memory**.

* It is used to refer the immediate super class as shown below

```
A
super; //Object class member
B
super; //A class member
C
super; //This super will refer to member of B class i.e immediate super
class, from here we cannot access the member of class A
```

* Just like this keyword, super keyword also represents non static members so, we cannot use super from static area like static method, static block or static nested class.

Super keyword we can use 3 ways in Java :

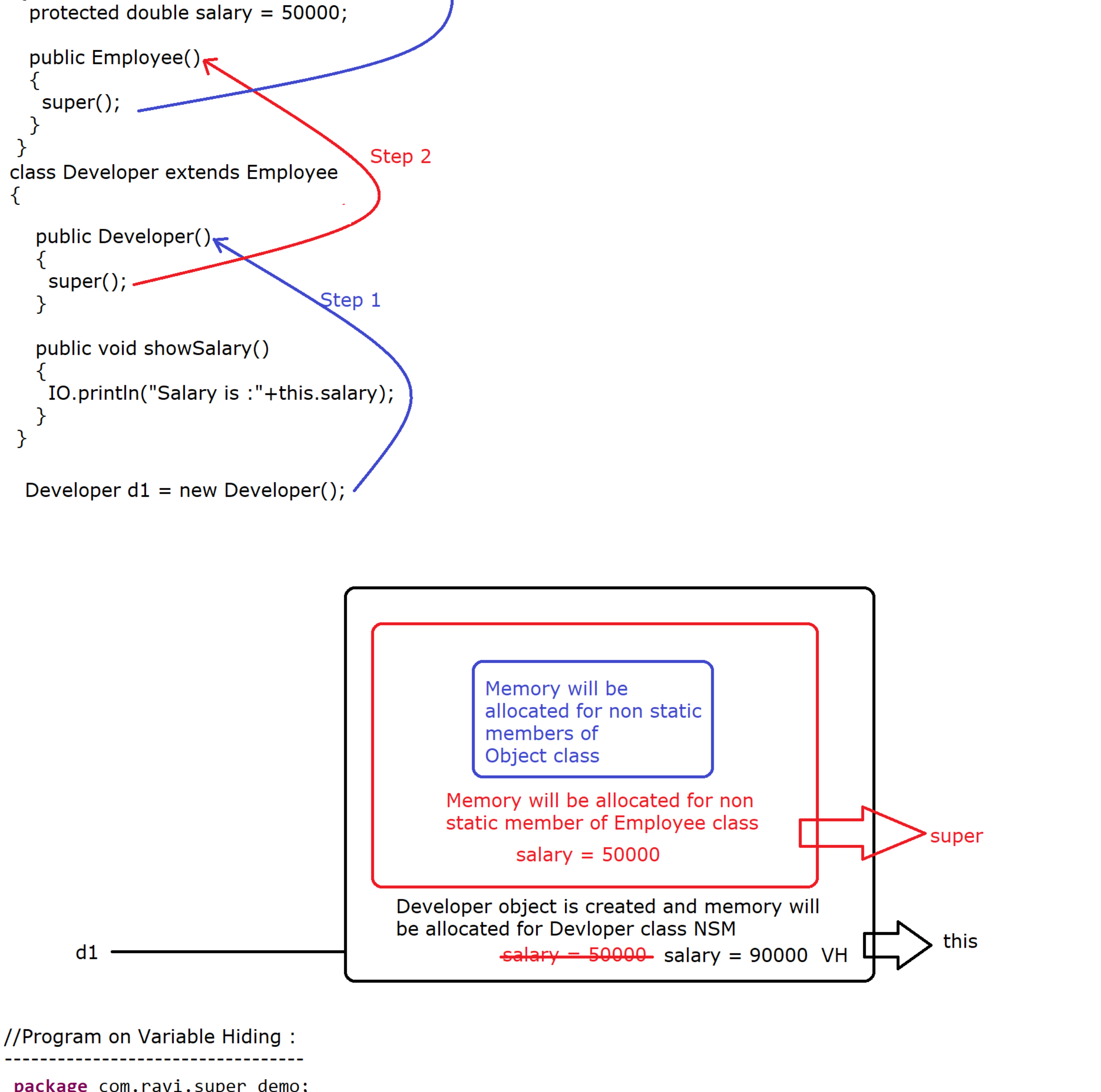
- 1) To access the non static field of super class (Variable Hiding)
- 2) To access the non static method of super class (With Method Overriding)
- 3) To access the constructor of super class (Constructor Chaining)

1) To access the non static field of super class (Variable Hiding)

Whenever super class non static field name and sub class non static field name both are exactly same then it is called variable Hiding, Here sub class non static field hides super class non static field. [shown in the diagram]

In order to access super class non static field i.e super class memory, we should use super keyword as shown in the program.

Note : Variable Hiding is only possible in super & sub class, on the other hand Variable shadowing is possible in the same class.



//Program on Variable Hiding :

```
package com.ravi.super_demo;

class Employee
{
    protected double salary = 50000;
}

class Developer extends Employee
{
    protected double salary = 90000; //Variable Hiding

    public void showSalary()
    {
        IO.println("Employee Salary :"+super.salary);
        IO.println("Developer Salary is :"+this.salary);
    }
}

public class VariableHidingDemo
{
    public static void main(String[] args)
    {
        Developer d1 = new Developer();
        d1.showSalary();
    }
}
```

2) To access super class non static method (Method Overriding)

If the super class non static method name and sub class non static method name both are same (Method Overriding) and if we create an object for sub class then sub class method will be executed (bottom to top), if we want to call super class method from sub class method body then we should use super keyword as shown in the program.

//Programs :

```
package com.ravi.super_demo;

class Bird
{
    public void fly()
    {
        IO.println("Bird is flying");
    }
}

class Parrot extends Bird
{
    public void fly()
    {
        super.fly();
        IO.println("Parrot Bird is flying");
    }
}

public class MethodOverridingDemo1
{
    public static void main(String[] args)
    {
        Parrot p = new Parrot();
        p.fly();
    }
}

package com.ravi.super_demo;

class Alpha
{
    public String toString()
    {
        return "Alpha";
    }
}

class Beta extends Alpha
{
    public String toString()
    {
        IO.println(super.toString());
        return "Beta";
    }
}

class Gamma extends Beta
{
    public String toString()
    {
        IO.println(super.toString());
        return "Gamma";
    }
}

public class MethodOverridingDemo2
{
    public static void main(String[] args)
    {
        Gamma g = new Gamma();
        IO.println(g);
    }
}
```